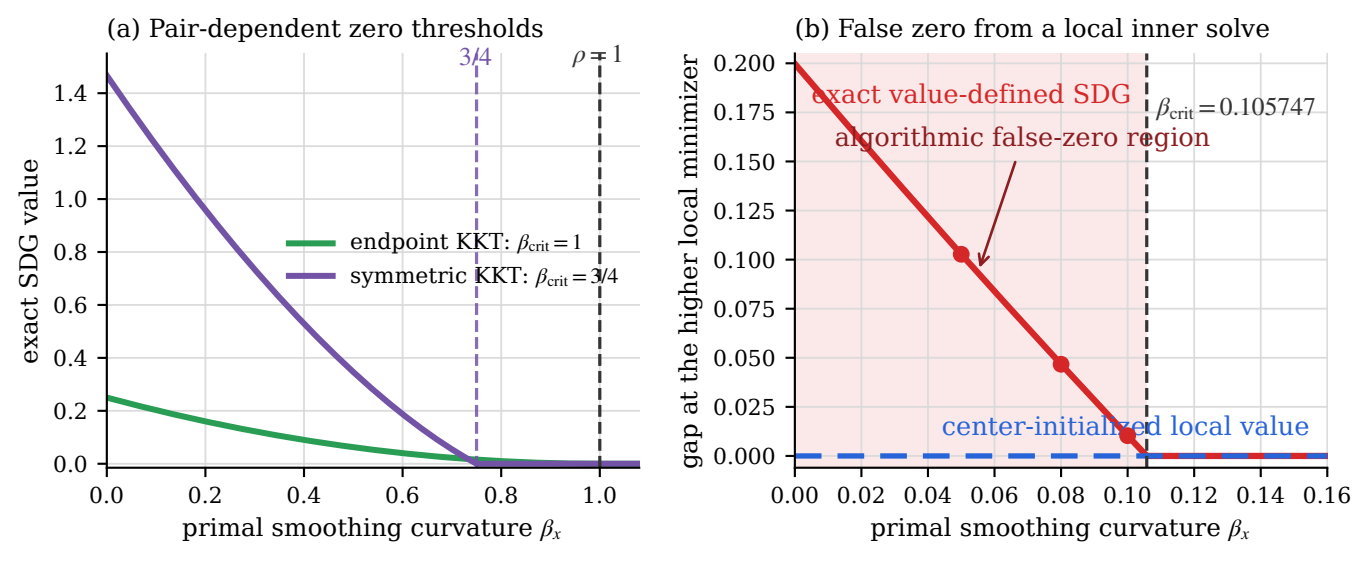
(a) Pair-dependent zero thresholds
3/4
$\rho = 1$
endpoint KKT: $\beta_{\mathrm{crit}} = 1$
symmetric KKT: $\beta_{\mathrm{crit}} = 3/4$
exact SDG value
primal smoothing curvature β_x
(b) False zero from a local inner solve
exact value-defined SDG
algorithmic false-zero region
$\beta_{\mathrm{crit}} = 0.105747$
center-initialized local value
gap at the higher local minimizer
primal smoothing curvature β_x